

Image Analysis Report

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Executive Summary

This cross-junction assembly exhibits severe atmospheric corrosion (Level 4) characterized by heavy oxide scaling and total coating system failure. The structural integrity of the flange fasteners and the pressure-retaining wall of the fitting are significantly compromised and require immediate assessment.

Key Findings

#	Finding
1	Complete breakdown of the protective coating system with extensive peeling, blistering, and delamination across all visible surfaces.
2	Formation of heavy, loose oxide scale on the central 4-way fitting, indicating advanced material loss.
3	Severe corrosion on flange bolts and nuts, with visible rounding of bolt heads and deep oxidation that may prevent safe removal.
4	Active crevice corrosion visible at the flange-to-flange interfaces, potentially compromising the gasket seal.
5	Presence of bright orange/red rust indicates an active, ongoing corrosion process in a high-moisture or salt-laden environment.
6	Localized pitting is highly probable beneath the stratified scale layers on the central fitting body.

7	Specific wall thickness loss cannot be determined from the image and requires quantitative NDT.
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Localized detections (VLM, approximate)

- Heavy Scaling (conf. 95%): x=0.270, y=0.240, w=0.450, h=0.620
- Coating Failure (conf. 92%): x=0.010, y=0.340, w=0.170, h=0.380
- Coating Failure (conf. 92%): x=0.810, y=0.310, w=0.190, h=0.410
- Corroded Fasteners (conf. 88%): x=0.160, y=0.230, w=0.150, h=0.610
- Corroded Fasteners (conf. 88%): x=0.680, y=0.230, w=0.150, h=0.610
- Crevice Corrosion (conf. 85%): x=0.210, y=0.230, w=0.050, h=0.610

Recommendations

#	Recommendation
1	Schedule Ultrasonic Testing (UT) for wall thickness measurements on the central fitting and pipe spools within 48-72 hours.
2	Isolate and depressurize the system before any mechanical intervention due to the risk of structural failure of the fasteners.
3	Replace 100% of the flange bolting hardware; existing fasteners are unsuitable for continued service or re-torquing.
4	Abrasive blast clean the entire assembly to Sa 2.5 (Near-White Metal) to expose the metal substrate for a full integrity inspection.
5	Perform Magnetic Particle Inspection (MPI) or Dye Penetrant Inspection (DPI) on the fitting welds to check for stress corrosion cracking.
6	Re-apply a high-performance three-coat maintenance coating system (Zinc-rich primer, Epoxy intermediate, Polyurethane topcoat).
7	Increase inspection frequency to 6-month intervals until the environmental causes of rapid degradation are mitigated.